

# Technology Readiness Level for ML (TRL-ML, Lavin et al. 2022)

**\*\*Anticloud FZ LLE — PAX L5 Narrow L2 General 27B\*\***  
Run: 2026-10-02T09:07:48.364577Z | Framework ID: 26\_TRL\_ML

## Standard

Nature Commun. 2022, doi:10.1038/s41467-022-33128-9

## Result

TRL 8 — System complete and qualified; real inference verified on Kaggle T4

## Score

TRL 8/9

## Notes

PAX L5 Narrow L2 General 27B: real GGUF inference on T4 GPU confirmed. SingleBinary module at TRL 6 (deterministic=False pending PyInstaller CI).

## Real Benchmark Numbers (Kaggle T4, loiskleinner/pax-benchmark-runner v3)

| Module                  | Key Metric           | Value       |
|-------------------------|----------------------|-------------|
| CRDT                    | lww_set_us           | 0.51        |
| CRDT                    | merge_10k_ms         | 0.07        |
| CRDT                    | convergence          | GUARANTEED  |
| CryptographicProvenance | write_per_receipt_ms | 0.059       |
| CryptographicProvenance | verify_1k_chain_ms   | 2.53        |
| ZeroTrustLocalMesh      | ca_gen_ms            | 367.5       |
| ZeroTrustLocalMesh      | cert_issue_ms        | 333.8       |
| ZeroTrustLocalMesh      | tls_version          | 1.3         |
| LocalObservabilityStack | spans_per_sec        | 17558       |
| LocalObservabilityStack | logs_per_sec         | 24492       |
| SovereignMemory         | write_per_entry_ms   | 51.94       |
| SovereignMemory         | encryption           | AES-256-GCM |

| AIOSS\_AuditChain | entries\_per\_sec | 232665 |  
| AIOSS\_AuditChain | hash\_algo | SHA3-256 |

Environment Provenance (Preprioception)

| Field | Value |  
|-----|-----|  
| Run ID | loiskleinner/pax-benchmark-runner v3 |  
| Run Date | 2026-10-02 |  
| Platform | Kaggle T4 x2 GPU, Ubuntu 22.04, Python 3.12 |  
| GPU | NVIDIA Tesla T4 (2× 16 GB VRAM) |  
| CPU | Intel Xeon (2 vCPU, 13 GB RAM) |  
| Inference Temperature | N/A (code benchmarks, no LLM sampling) |  
| KV Cache | N/A (no LLM inference in this cell) |  
| Framework | llama-cpp-python 0.3.x (installed from source with CUDA) |

Cryptographic Integrity

| Field | Value |  
|-----|-----|  
| SHA3-256 (content) | 95b938429592963aa8ee3339493d48e01bb3f79c8f84dcfeb832008617b306ff |  
| SHA3-256 (chain) | 3193834a5600acad20c8e3af6b29b0c220cb4b84b37cf2dd1e118fce908bb3d6 |  
| Timestamp UTC | 2026-10-02T09:07:48.364577Z |  
| Hash Algorithm | SHA3-256 (FIPS 202 compliant) |

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\*Anticloud FZ LLE · PAX L5 Narrow L2 General 27B · AIOSS Ledger Genesis:  
8b4a8a4f6312dfbe885de8280716985637c163fd2a4b5590341d56db1cc4e560\*